King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 241

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $y = \dfrac{(x^2+2)(x^3+3)}{(x^2-2)(x^3-3)}$, then the value of $\dfrac{dy}{dx}$ at $x = 1$ is equal to

(a) $\dfrac{17}{2}$

(b) $\dfrac{59}{2}$

(c) $\dfrac{39}{4}$

(d) $\dfrac{49}{2}$

(e) $\dfrac{29}{4}$

2. If $y = (1+x^2)^{\cos x}$, then $\dfrac{y'}{y} =$

(a) $\dfrac{2x\cos x - \sin x \cdot (1+x^2)\ln(1+x^2)}{1+x^2}$

(b) $\dfrac{x\cos x + (1+x^2)\sin x}{1+x^2}$

(c) $\dfrac{2x\cos x - \sin x \cdot \ln(1+x^2)}{1+x^2}$

(d) $-\dfrac{2x\sin x}{1+x^2}$

(e) $\dfrac{\cos x - \sin x}{1+x^2}$

3. If m and M are respectively the absolute minimum and absolute maximum values of the function $f(x) = x^3 - \dfrac{3}{2}x^2 + 1$ on the interval $[-1, 2]$, then $m + M =$

(a) $-\dfrac{1}{2}$

(b) -1

(c) 0

(d) $\dfrac{3}{2}$

(e) $\dfrac{7}{2}$

4. If (A, B) and (C, D) are the points on the graph of the function $f(x) = \dfrac{x+1}{x-1}$ at which the tangent lines are perpendicular to the line $2x - y = 1$, then $A + B + C + D =$

(a) 5

(b) 4

(c) -1

(d) 3

(e) 0

5. The slope of the tangent line to the graph of $f(x) = \dfrac{4x}{x^2+6}$ at the point $\left(2, \dfrac{4}{5}\right)$ is equal to

(a) $\dfrac{1}{25}$

(b) $\dfrac{2}{25}$

(c) $\dfrac{4}{25}$

(d) 0

(e) $\dfrac{3}{25}$

6. If $y = Ax + B$ is an equation of the tangent line to the graph of $f(x) = 1 + x\cos x$ at the point $(0, 1)$, then $AB =$

(a) -1

(b) 0

(c) 1

(d) 2

(e) $-\dfrac{1}{2}$

7. If $f(x) = \dfrac{1}{e^x}$, then $f^{(55)}(0) =$

(a) $\dfrac{1}{e}$

(b) e

(c) -1

(d) $-\dfrac{1}{e}$

(e) 1

8. The value of c that satisfies the Mean Value Theorem when applied to $f(x) = 2x - \ln x$ on the interval $[1, e]$ is equal to

(a) $\dfrac{e+1}{e}$

(b) $2e - 3$

(c) $e - 1$

(d) $\dfrac{2e}{e-1}$

(e) $e - \dfrac{1}{2}$

9. Let f be a differentiable function and

$$g(x) = f\left(x^3 - 3\sin(\pi x)\right), \quad f(1) = 3, \quad f'(1) = \frac{1}{3}.$$

Then $g'(1) =$

(a) $\pi + 1$

(b) π

(c) $1 - 3\pi$

(d) $3 - 3\pi$

(e) $3\pi + 3$

10. If $f(x) = x\sqrt{x-3}$, then $(f^{-1})'(4) =$ (Hint: What is the value of $f(4)$?)

(a) $\dfrac{1}{4}$

(b) $\dfrac{1}{5}$

(c) $\dfrac{1}{3}$

(d) $\dfrac{1}{2}$

(e) 1

11. If $x^2 + y^2 - 3\ln y = 10$, then, by implicit differentiation, $\dfrac{dy}{dx} =$

(a) $\dfrac{3xy}{2-3y^2}$

(b) $\dfrac{2x}{3-y^2}$

(c) $\dfrac{xy}{1-2y^2}$

(d) $\dfrac{2xy}{3-2y}$

(e) $\dfrac{2xy}{3-2y^2}$

12. Using Newton's Method to approximate the zero of the function $f(x) = x - 2\sqrt{x+1}$, starting with $x_1 = 3$, we get $x_2 =$

(a) $\dfrac{7}{2}$

(b) $\dfrac{9}{2}$

(c) 5

(d) 4

(e) $\dfrac{5}{2}$

13. Air is being pumped into a spherical balloon at a rate of 32π cm^3/min. When the volume of the balloon is $\dfrac{32\pi}{3}$ cm^3, the radius of the balloon is changing at a rate of (Hint: The volume of the sphere is $V = \dfrac{4}{3}\pi r^3$)

(a) 4 cm/min

(b) $\dfrac{3}{2}$ cm/min

(c) 3 cm/min

(d) $\dfrac{3}{4}$ cm/min

(e) 2 cm/min

14. If $y = x^2\sqrt{4-x^2}$, then $\dfrac{dy}{dx} =$

(a) $\dfrac{4x+3x^3}{\sqrt{4-x^2}}$

(b) $\dfrac{-x^3}{\sqrt{4-x^2}}$

(c) $\dfrac{2x-x^3}{\sqrt{4-x^2}}$

(d) $\dfrac{8x+x^3}{\sqrt{4-x^2}}$

(e) $\dfrac{8x-3x^3}{\sqrt{4-x^2}}$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$y(1) = 6$, $\frac{y'}{y} = \frac{59}{12}$, so $y'(1) = \frac{59}{2}$.
2	**(a)**	$\ln y = \cos x \ln(1 + x^2)$, so $\frac{y'}{y} = \frac{2x \cos x}{1+x^2} - \sin x \ln(1 + x^2)$.
3	**(d)**	Values $1, \frac{1}{2}, -\frac{3}{2}, 3$: $m + M = -\frac{3}{2} + 3 = \frac{3}{2}$.
4	**(b)**	$f' = -\frac{1}{2}$ gives $x = 3, -1$; points $(3, 2), (-1, 0)$; sum $= 4$.
5	**(b)**	$f'(x) = \frac{24-4x^2}{(x^2+6)^2}$, so $f'(2) = \frac{2}{25}$.
6	**(c)**	$f'(0) = 1$, tangent $y = x + 1$, so $AB = 1$.
7	**(c)**	$f^{(n)}(x) = (-1)^n e^{-x}$, so $f^{(55)}(0) = -1$.
8	**(c)**	$2 - \frac{1}{c} = \frac{2e-3}{e-1}$ gives $c = e - 1$.
9	**(a)**	$g'(1) = f'(1)(3 + 3\pi) = \pi + 1$.
10	**(c)**	$f(4) = 4$, $f'(4) = 3$, so $(f^{-1})'(4) = \frac{1}{3}$.
11	**(e)**	$2x + 2yy' - \frac{3}{y}y' = 0$, so $y' = \frac{2xy}{3-2y^2}$.
12	**(c)**	$f(3) = -1$, $f'(3) = \frac{1}{2}$, $x_2 = 3 + 2 = 5$.
13	**(e)**	$r = 2$; $32\pi = 4\pi r^2 r'$ gives $r' = 2$ cm/min.
14	**(e)**	Product and chain rules: $y' = \frac{8x-3x^3}{\sqrt{4-x^2}}$.